

Week 4 Project Report

DeployMate: DevOps Deployment Troubleshooter RAG

Problem Statement

Beginners often get stuck when moving from localhost to deployment. DeployMate helps by retrieving relevant DevOps runbook entries first, then generating a grounded answer using AI.

Option Chosen

Coding + AI API -> RAG

Tools and Technologies

- Node.js web app
- OpenAI API support for embeddings and generation
- RAG over a DevOps runbook
- Dockerfile for containerization
- GitHub Actions workflow for CI

How the RAG Flow Works

Step	What Happens
1	The app loads DevOps notes from data/devops_runbook.json.
2	The notes are chunked and embedded.
3	The user asks a deployment question.
4	Relevant chunks are retrieved using similarity search.
5	AI generates an answer only from retrieved sources.
6	The UI shows the answer and source snippets.

Sample Input and Output

Input	Output
My Node app works on localhost but not on AWS public IP. What should I check?	Check AWS security group, public IP testing, host binding, exposed port, environment variables, and firewall rules. The source snippets explain AWS EC2 deployment and localhost-to-public debugging.

Demo Video Script

- Introduce DeployMate as a Week 4 DevOps RAG project.
- Explain the deployment troubleshooting problem.
- Ask a sample question in the web UI.
- Show the generated answer and source citations.
- Explain that retrieval happens before generation.
- Show Dockerfile and GitHub Actions files.

Submission Notes

Upload the complete week4-devops-rag-troubleshooter folder as a ZIP or GitHub repository. Do not upload .env because it may contain an API key.